MIL-S-47053(MI)
10 May 1974
SUPERSEDING
MIS-13962
8 November 1967

MILITARY SPECIFICATION

SLEEVING, TEXTILE, BRAIDED, POLYAMIDE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of tubular braided, black polyamide sleeving. The sleeving is suitable for use within a temperature range of minus 55 degrees Celsius (C) (minus 67 degrees Fahrenheit (F) to 105 degrees C (221 degrees F).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-572	Cord, Yarns and Monofilaments Organic Synthetic Fiber
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

FSC 5970

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MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall meet all the requirements of this specification.

3.2 Material.

3.2.1 Composition and construction.

3.2.1.1 Composition. The sleeving shall be braided from 210 denier yarn conforming to MIL-C-572, Type P, Form Y.

3.2.1.2 Construction. Sleeving construction shall be as specified below and in Table I.

a. Yarn. Yarn used in the sleeving braid shall be 2 ply with an "S" twist (see 6.3).

b. Cordage. Cordage used in the sleeving braid shall be 2 ply, 840 denier, with a "Z" twist (see 6.3).

c. Spliced joints. Spliced joints in the braided sleeving shall be limited to fused joints in yarn or cordage only. Spliced joints shall not appear more often than once in 25 yards of sleeving, or not more often than 20 times in 600 yards.

3.2.2 Sleeving dimensions. Sleeving dimensions shall be as specified in Table I.

3.2.3 Color. Unless otherwise specified, the sleeving color shall be gloss black using a chrome fast dye.

3.2.4 Braid coverage. The coverage of the braid shall not be less than 85 percent.

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3.3 Performance characteristics.

3.3.1 Tensile strength (tenacity). The breaking strength of the braid cord shall not be less than 25 pounds. The breaking strength of fused cord splices shall not be less than 70 percent of the cord breaking strength.

Table I. Sleeving Dimensions and Construction

Inside diameter, inch, nominal	Number of ends	Number of carriers	Picks per inch	Wall thickness, inch
1/8	24	24	20 ± 2	0.030 ± 0.015
3/16	32	32	18 ± 2	
1/4	44	44	20 ± 2	
5/16	44	44	18 ± 2	
3/8	48	48	18 ± 2	
1/2	64	64	16 ± 2	
5/8	128	64	12 ± 2	0.040 ± 0.015
3/4	128	64	11 ± 2	
7/8	192	64	10 ± 2	
1	216	72	10 ± 1	0.050 ± 0.015
1-1/4	216	72	9 ± 1	
1-1/2	288	72	8 ± 1	0.060 ± 0.015
1-3/4	288	72	7 ± 1	
2	360	72	6 ± 1	

3.4 Environmental conditions.

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3.4.1 Fungus resistance. The sleeving shall not support the growth of fungus.

3.5 Workmanship. The sleeving shall be free of visible defects, dirt, grime, grease, and abraded or damaged surfaces.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. Unless otherwise specified in the contractual document (see 6.2), a sample consisting of 20 feet of sleeving, manufactured under the same conditions as those proposed for subsequent production, shall be subjected to preproduction inspection. Accomplishment of preproduction inspection shall be as specified herein. Subsequent units will not be considered for acceptance until Government approval of the preproduction sample has been obtained. Units subjected to preproduction inspection shall have successfully passed acceptance inspection. Testing of the preproduction sample, to determine compliance with the characteristics listed in Table II, shall be conducted in accordance with the corresponding test paragraphs.

Table II. Preproduction Inspection

Characteristic	Requirement source	Test paragraph
Tensile strength	3.3.1	4.5.4.5
Fungus resistance	3.4.1	4.5.5.1

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4.3 Quality conformance inspection.

4.3.1 Sampling for acceptance inspection. Sampling for acceptance inspection shall be conducted in accordance with MIL-STD-105, at an acceptable quality level (AQL) of 4.0 percent defective.

4.3.2 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.3.3 Acceptance inspection. Acceptance inspection of the sample specified in 4.3.1, to determine compliance with the characteristics specified in Table III, shall be conducted in accordance with the corresponding test and inspection paragraphs.

4.4 Inspection equipment. The inspection equipment for conducting examination and tests shall be as specified in the applicable test methods and procedures paragraphs.

Table III. Acceptance Inspection

Characteristic	Requirement source	Test and inspection paragraph
Composition and construction	3.2.1	4.5.4.1
Sleeving dimensions	3.2.2	4.5.4.2
Color	3.2.3	4.5.4.3
Braid coverage	3.2.4	4.5.4.4
Workmanship	3.5	4.7
Packaging and packing	5.1	4.6
Marking	5.2	4.6

4.5 Test methods and procedures.

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4.5.1 Test conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements:

a. Temperature, room ambient (16 to 32 degrees C) (60 to 90 degrees F)).

b. Altitude, facility ground.

c. Humidity, facility ambient up to 95 percent relative humidity.

4.5.2 Test sequence. Test sequence within each classification of inspection (preproduction or acceptance) shall be at the option of the supplier.

4.5.3 Test specimen preparation. Test specimens shall be cut from the sample specified in 4.3.1.

4.5.4 Tests.

4.5.4.1 Composition and construction. The composition shall be certified by the supplier and the construction shall be determined visually to establish conformance to 3.2.1.

4.5.4.2 Dimensions. The sleeving wall thickness shall be measured with a paper micrometer or equivalent instrument for compliance with 3.2.2.

4.5.4.3 Color. The sleeving shall be visually examined for compliance with color requirements in 3.2.3.

4.5.4.4 Braid coverage. Three specimens of sleeving, each not less than 6 inches in length, shall be carefully cut and fitted over a smooth surfaced mandrel which has hemispherical ends. The mandrel diameter shall correspond in size to the inside diameter of the sleeving within plus and minus 1/64 inch. The sleeving shall be fitted on the mandrel in a manner where it is not stretched (tightly fitted), loosely fitted, compressed, or gathered. The braid coverage shall then be determined by using the formula below and shall conform to 3.2.4. Applicable values to be substituted in the formula shall be carefully obtained from the mandrel mounted sleeving specimen and the braid coverage computed.

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Braid coverage formula:

$$K = (2F - F^2)\ 100,$$

where K = percent coverage

F = NP x 0.0234/sine a

a = angle of braid with longitudinal axis of sleeving

$$\text{Tangent } a = 2\ (D + 0.0468)\ P/C$$

N = number of ends per carrier

P = picks per inch of sleeving length

D = inside diameter of sleeving in inches

C = number of carriers

4.5.4.5 Tensile strength (tenacity). Tensile strength of cord, carefully removed from finished sleeving, shall be conducted in accordance with applicable test procedures of MIL-C-572 to establish compliance with 3.3.1.

4.5.5 Environmental test.

4.5.5.1 Fungus resistance. Testing for fungus resistance shall be in accordance with applicable test procedures of MIL-E-5272 to establish conformance to 3.4.1.

4.6 Packaging, packing and marking. The inspector shall ascertain that the packaging, packing, and marking of the material conforms to this specification.

4.7 Workmanship. The sleeving shall be visually examined for conformance to workmanship requirements specified in 3.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. The sleeving shall be packaged and packed in accordance with standard industry practice. Unless otherwise specified in the contract or order, the sleeving shall be separated by size and furnished on spools or reels.

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5.2 Marking. Internal and external packaging shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name and address.

b. Manufacturer's designation.

c. Lot number or batch number.

d. Date of manufacture.

e. Number of this specification.

f. Sleeving size and quantity.

6. NOTES

6.1 Intended use. The sleeving in accordance with this specification is intended for use as protective sheathing over internal wire and cable bundles and harnesses in electronic equipment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required (see 3.1), and if so, pertinent details.

c. Quantity and size of sleeving.

d. Requirement for certification of composition (see 4.5.4.1).

e. Whether sleeving color, other than black, is required (see 3.2.3).

6.3 Definitions. Definitions specified in MIL-C-572 are applicable to corresponding terminology in this specification.

6.4 Supersession data. This specification includes the requirements of Missile Interim Specification, MIS-13962, dated 8 November 1967.

Custodian:
Army-MI

Preparing Activity:
Army-MI

Project No. 5970-A461

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-47053(MI)	SLEEVING, TEXTILE, BRAIDED, POLYAMIDE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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